COMBINATIONAL LOGIC DESIGN

Design and Simulation of an Automatic Street Light Control System

Intelligent, energy-efficient street lighting control using minimized logic gates

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01. CONTEXT & GOALS

Introduction and Project Objectives

Autonomous Control

Automated switching based on environmental factors without manual intervention.

Energy Efficiency

Eliminates daytime operation and minimizes unnecessary off-peak night usage.

Combinational Logic

Instantaneous switching using pure logic gates without clocks or memory elements.

Core Rule: Lights active strictly at night when peak traffic occurs or motion is detected.

02. LOGIC CRITERIA

Environmental Decision Rules

L = 0 (Day)

Daytime Override

Lights remain OFF during daylight regardless of motion or time.

L = 1, T = 1

Night Peak Hours

Lights ON for continuous safety during high traffic peak hours.

L = 1, T = 0

Night Off-Peak

Lights ON only when motion is detected (M = 1) to conserve power.

03. OPTIMIZATION

Boolean Expression & K-Map Simplification

L \ MT	00	01	11	10
L = 0	0	0	0	0
L = 1	0	1	1	1

SOP Expression

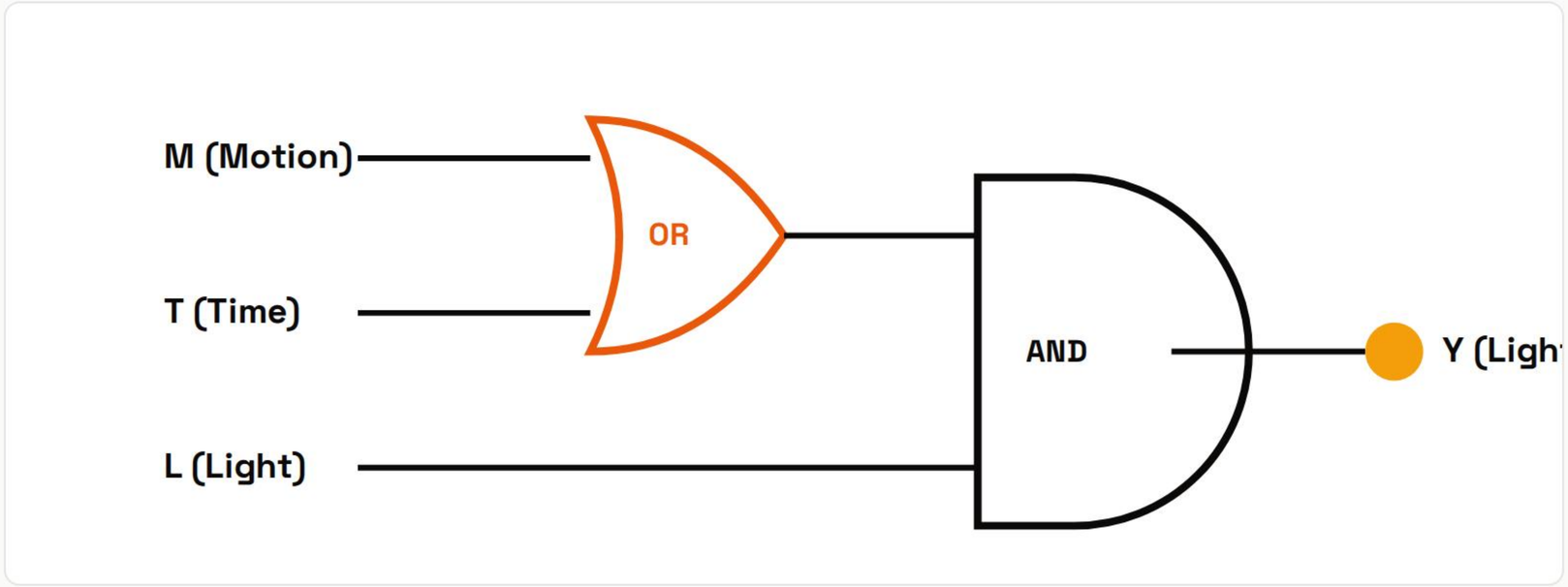
$$Y = L \cdot T + L \cdot M$$

Factored Minimal Form

$$Y = L \cdot (T + M)$$

04. GATE ARCHITECTURE

Minimized Logic Circuit Diagram



Gate 1: 2-Input OR combining Time (T) and Motion (M)

Gate 2: 2-Input AND enabling output only during Night (L=1)

05. LOGISIM SIMULATION

Circuit Design & Workflow

STEP 1

Pin Assignment

Set input toggle switches for Light (L), Motion (M), and Time (T).

STEP 2

Gate Wiring

Wire one OR gate and one AND gate to construct Y = L·(T + M).

STEP 3

Output Probe

Attach LED indicator to probe output Y for visual state verification.

06. VERIFICATION

Full System Truth Table

L (Light)	M (Motion)	T (Time)	Output (Y)	Operational Status
0	0	0	0	Daylight — Lights OFF
0	0	1	0	Daylight — Lights OFF
0	1	0	0	Daylight — Lights OFF
0	1	1	0	Daylight — Lights OFF
1	0	0	0	Night, Off-Peak, No Motion — Power Saving
1	0	1	1	**Night, Peak Hours — Lights ON**
1	1	0	1	**Night, Off-Peak, Motion — Lights ON**
1	1	1	1	**Night, Peak, Motion — Lights ON**

07. LOGISIM OUTPUT

Simulation Results & Analysis

Active States

Simulation triggers LED activation exclusively for input vectors 101, 110, and 111.

Hardware Reliability

Zero false positives during simulated daylight conditions across all 4 daytime states.

08. DEPLOYMENT

Real-World Applications

Smart Cities

Urban street grid automation to reduce municipal electricity expenditure.

Highways & Expressways

Traffic-responsive lighting during late-night hours on low-density roads.

Campus & Parks

Motion-triggered lighting for university pathways and private complexes.

09. FUTURE SCOPE

Possible Enhancements

PWM Dimming Control

Transition from binary ON/OFF to variable light intensities (e.g., 30% ambient to 100% on motion).

IoT Telemetry

Integrate microcontrollers for remote fault reporting, energy monitoring, and cloud telemetry.

10. HARDWARE INTERFACE

Real World Sensor Inputs

LDR Sensor (Light Input)

Detects ambient daylight for automatic dusk/dawn threshold switching (L).

PIR Sensor (Motion Input)

Detects pedestrian and vehicular movement during off-peak hours (M).

RTC Module (Time Input)

Schedules peak traffic intervals for guaranteed night illumination (T).

11. SUMMARY

Project Summary & Takeaways

50%

Hardware Reduction

K-Map factoring reduced the design to just 2 logic gates.

100%

Truth Table Compliance

Verified across all 8 possible combinational logic input states in Logisim.

PRESENTATION CONCLUDED

Thank You!

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